

Deepak Mallampati

AI/ML ENGINEER - AWS & GENERATIVE AI

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Kent, OH | US-based, F-1 OPT

PROFESSIONAL SUMMARY

AI/ML engineer with 4+ years building scalable AI services and data pipelines on AWS (Bedrock, SageMaker, Lambda). Strong across RAG (dense retrieval + cross-encoder re-ranking), NLP (BERT/RoBERTa NER at 89% F1), and ETL automation (60% fewer failed runs). Builds LLM and prompt-engineering solutions for production use cases including summarization, translation, and Text2SQL.

SKILLS

AWS (Bedrock/SageMaker/Lambda) • RAG & Retrieval • NLP & NER • Python ETL Pipelines • Prompt Engineering • Model Deployment & Monitoring

Generative AI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG | Prompt Engineering | Multi-Agent Systems | Fine-tuning (LoRA/QLoRA/PEFT) | Hugging Face

ML & NLP: PyTorch, TensorFlow, Keras | NER, Classification, Summarization | RAGAS, BERTScore | Drift Detection | A/B Testing

Vector Search: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching

MLOps & Cloud: AWS (Lambda, Bedrock, SageMaker), Azure OpenAI, GCP Vertex | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B | Observability

Data: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git, Jira

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously, cutting token consumption by **42%** while holding task accuracy across multi-step workflows.
- Engineered **retrieval-augmented generation (RAG)** pipelines over large healthcare corpora, pairing dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination on domain queries.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retries, reducing failed production runs by **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency and accuracy dashboards to catch regressions before release.
- Cut inference cost and latency via prompt compression, semantic caching, and model routing while holding SLA targets.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing model-selection decisions.
- Engineered content controls for policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining quality.
- Built **12 APIs and microservices**, cutting p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered across product, platform, and data engineering to deliver LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on clinical hospital-readmission data (k-anonymity, l-diversity, Laplace differential privacy).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (RF AUC 0.689 vs 0.696).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an H100 for controllable long-form generation; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Trained supervised **regression and classification models** on operational pipeline-storage data for failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, reaching **89% F1** on custom NER tasks.

PROJECTS

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites**, turning a multi-step manual translation chore into a single inline action.

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility tradeoff.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0